Phase 1A — Discovery & Core Design: foxotale

Date: 2026-09-28
Status: Approved
Sources: `.fsd3/foxotale/sd.md`, `.fsd3/foxotale/epic-breakdown.md`, Phase 1A discovery answers
Workflow: FSD3 Full · Phase 1 MVP only

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1. Problem & Context

FoxOtale needs paired **Flutter** mobile apps (Guest + Staff) backed by a **Python/FastAPI** API on **AWS + PostgreSQL**, so partner hotels take **direct bookings**, keep vouchers/bills/GST invoices in one guest system of record, and run property setup/ops/intelligence without OTA/tool sprawl.

In scope (MVP): SD Phase 1 — Guest discover→book→pay→documents; Staff RBAC, setup, reservations, rooms/tasks, request-queue contract, manual intelligence.
Out of scope: SD Phase 2–3 roadmap (in-stay depth, PMS auto-feeds, loyalty, channel manager, etc.).

Success (from SD): Direct-booking share uplift; checkout completion ≥70%; returning book <4 min; staff request/task ops clarity.

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2. Codebase Conventions (Greenfield)

Area	Finding
Workspace	No FoxOtale application source yet (`foxOtale/` = PRDs only). Unrelated CoreAI templates ignored.
Clients	Flutter Guest + Staff apps (Material 3 + brand theme) — per SD / UX ideation
API	**Python + FastAPI** (confirmed)
Data	**PostgreSQL on AWS** (confirmed)
Tests (API)	**pytest** + httpx `AsyncClient` (recommended default)
Packaging	**uv** or Poetry (recommend **uv**)
Migrations	**Alembic**
CI	To be established in 1C (SAST/SCA/secret scan recommended: Bandit/Ruff, pip-audit, gitleaks)

Established patterns to introduce (no legacy to match):

- Modular monolith FastAPI: `routers → services → repositories`
- SQLAlchemy 2.0 async + Alembic
- JWT (guest) + JWT/staff session with RBAC claims
- OpenAPI-first contracts (Phase 1B)

- Shared Flutter packages: `foxotale_ui`, `foxotale_domain`, `foxotale_api`

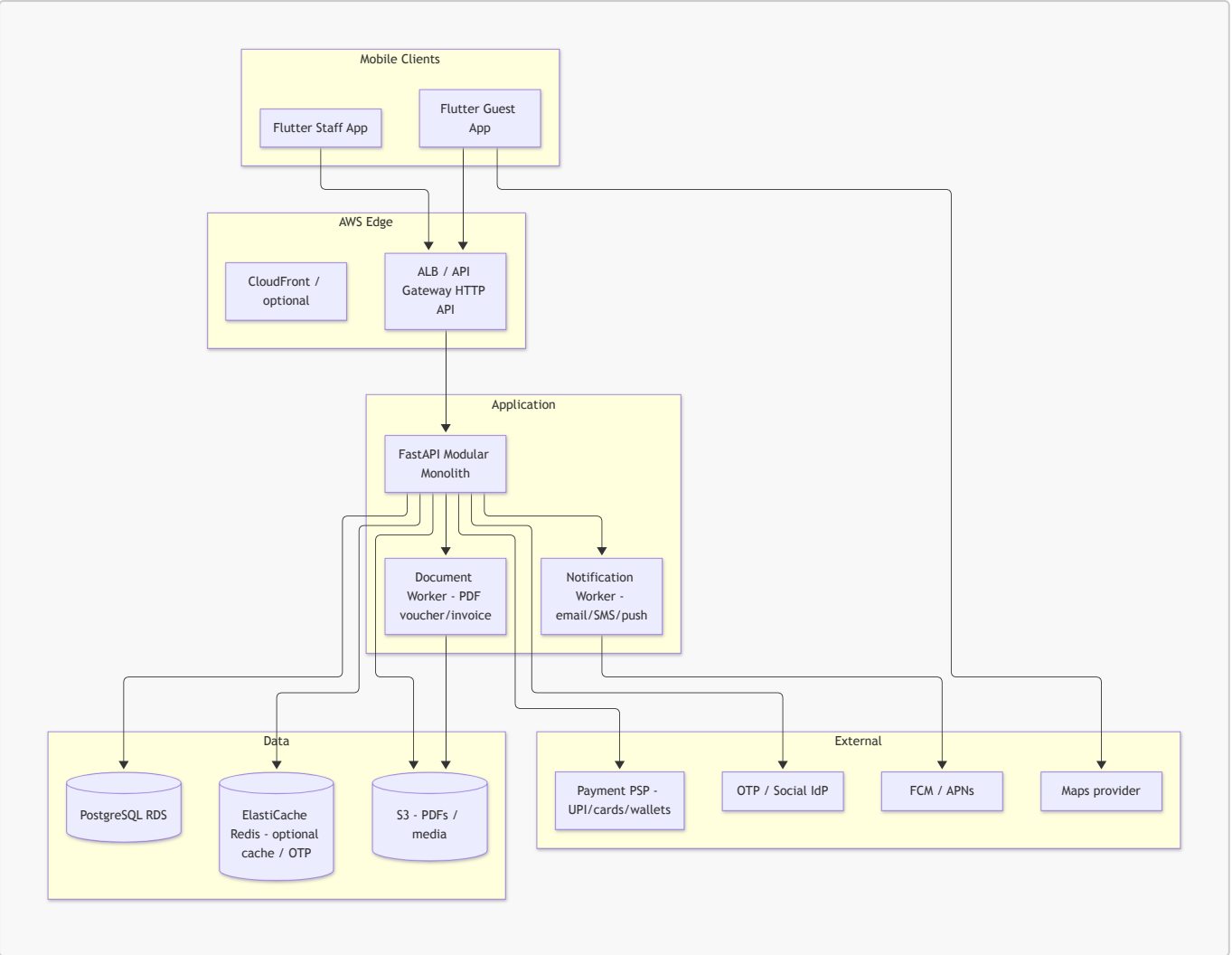
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3. Discovery Answers & Pilot Scale Recommendations

Topic	Decision
Backend	Python / FastAPI
Cloud + DB	AWS + PostgreSQL
Region	**Asia Pacific** — recommend primary **`ap-south-1` (Mumbai)** for India GST/UPI proximity; optional DR later in `ap-southeast-1`
Pilot properties	**Recommend 5–10** partner properties (mix of 1–2 standalone + one small chain of 3–5)
Peak concurrency	**Recommend design for ~500 concurrent guest sessions + ~100 concurrent staff** across the pilot (≈ tens of req/s steady; short spikes at checkout)
Inventory SoT (v1)	Staff App setup (manual rates/inventory) — PMS real-time deferred `[ASSUMPTION pending business]`
Intelligence data (v1)	Manual entry in Staff App

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4. Architecture Overview



Approaches considered

Dimension	A: Microservices per domain	B: Modular monolith FastAPI ✅	C: Serverless-only (Lambda + API GW)
Replaceability	Strong	Strong (ports/adapters)	Adequate
Cognitive Load	Weak	Strong	Adequate
Risk Isolation	Strong	Adequate	Adequate
Future Flexibility	Strong	Adequate (extract later)	Adequate
Security Posture	Adequate	Strong (fewer hops)	Adequate
Verdict	Overkill for pilot	**Chosen**	Harder local/dev + long-request PDF

ADR-001: Modular monolith on FastAPI + RDS Postgres

Status: Accepted
Date: 2026-09-28

Context: Greenfield MVP spanning Guest booking and Staff ops; pilot scale is modest; team needs one deployable API.

Decision: Single FastAPI service with domain modules (`identity`, `catalog`, `booking`, `documents`, `ops`, `intelligence`) sharing one Postgres database (schemas or table prefixes). Async workers (SQS + worker process or same process background tasks initially) for PDF and notifications.

Consequences:

- **Positive:** Fast delivery, simple ops, shared transactions for booking+voucher
- **Negative:** Single deploy unit; noisy-neighbour risk at scale
- **Risks:** God-module growth — enforce module boundaries in code review

Alternatives: Microservices (deferred); Firebase/Supabase (rejected — weaker fit for Staff RBAC + GST docs + relational reporting).

ADR-002: AWS region `ap-south-1`

Status: Accepted (pending infra confirmation)
Decision: Primary region Mumbai for APAC + India payment/GST latency.

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5. Core Data Model (entities)

```
Organization (chain optional)
├── id: uuid PK
├── name, type: standalone|chain
└── timestamps

Property
├── id, organization_id FK
├── name, status: draft|published
├── location (geo, city, country)
├── nearby_distances JSONB
└── facilities JSONB

RoomType / Room / RatePlan / MealPlan / InventoryCalendar / BlackoutDate
(Property-scoped; inventory counts + rates drive Guest availability)

GuestUser
├── id, phone/email, auth_providers
└── gst_profile JSONB, preferences

StaffUser
├── id, property_ids / area_ids scope
├── role: owner|area_gm|area_hm|gm|ops_mgr|supervisor|executive
└── one role per user (v1)

Booking
├── id, property_id, room_type_id, guest_user_id
├── check_in, check_out, guests_count, status
├── price_breakdown JSONB, gst_details JSONB
```

```
├── payment_status, amounts paid/pending
└── segment: upcoming|in_house|past (derived or stored)

Payment
├── id, booking_id, psp_ref, method, status, amount

Document
├── id, booking_id, type: voucher|bill|gst_invoice
├── s3_key, generated_at, metadata

Task / RoomStatus / GuestRequest
(Ops board; GuestRequest producers mostly Phase 2)

IntelligenceFact (manual v1)
├── property_id, period, metric_type, dimensions JSONB, values
(Supports YoY/MoM/budget/segment/guest-split/corporate — normalized later)
```

Relationships (key): Organization 1—N Property; Property 1—N RoomType/RatePlan/Booking; Booking 1—N Document/Payment; StaffUser N—M Property (scoped); Task N—1 Room/StaffUser.

Availability rule (v1): Search returns Property iff published AND inventory calendar has capacity for date range AND not blacked out AND rate exists — computed in catalog/booking services (no external channel manager).

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6. Security Context (discovery)

Topic	Phase 1 stance
Data classification	**PII** (guest phone/email/name); **confidential** financial (payments refs, GST); **PCI** = outsourced to PSP (no raw PAN in our DB)
Trust boundaries	App → TLS → ALB → FastAPI → RDS/S3/PSP/IdP
Authn Guest	Phone OTP + social/email; JWT access + refresh; browse anonymous
Authn Staff	Email/password or SSO later; JWT with role + property/area claims
Authz	**RBAC** strict ladder (SD §7 draft); property/area scoping; hide unauthorized UI
Threat actors	Anonymous scrapers, authenticated guests (IDOR), staff privilege escalation, dependency compromise
Top abuse	(1) IDOR on bookings/documents (2) Staff role elevation (3) Payment webhook forgery / replay
Standards	OWASP ASVS L2 target; India DPDP awareness for guest PII; GST invoice fields Legal-owned
CI security	Introduce Bandit/Ruff, pip-audit, gitleaks (no existing FoxOtale CI)

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7. Integration Touchpoints (MVP)

System	Purpose	Notes
PSP	UPI/cards/wallets	Provider TBD — Razorpay/Stripe India common for APAC/IN
OTP / Social	Guest auth	Twilio/MSG91 + Google/Apple as candidates
S3	Media + PDF storage	Private objects; signed URLs
FCM/APNs	Staff (and Guest) push	
Maps	Property location	Client-side SDK
Email	Confirmation + documents	SES

Document SoT: Prefer **API document service** triggered on booking confirm (voucher) and Staff check-out (final bill/GST) — single writer. `[ASSUMPTION — confirm vs PMS]`

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8. Performance & Scale Targets (pilot)

Metric	Target
Search p95	< 2s on typical mobile network path
Checkout confirm after PSP success	< 2s API side
Steady load	Design ~50 RPS headroom; pilot likely <20 RPS
DB	RDS Postgres db.t4g.medium starting point (right-size in 1C)
Growth	10× properties (→50–100) without re-architecture if modular boundaries held

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9. Open Questions (carry to 1B/1C)

#	Question	Owner
Q1	Exact PSP + webhook signing	Eng + Biz
Q2	Confirm document generation SoT (API vs PMS)	Product
Q3	Staff §7 matrix final sign-off	Product
Q4	Pay-at-hotel property flags	Biz
Q5	Cancellation policy global vs per-property	Legal
Q6	Redis required day-1 for OTP rate limits?	Eng (recommend yes lightweight)

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10. References

- `.fsd3/foxotale/sd.md` — approved Solution Design
- `.fsd3/foxotale/epic-breakdown.md` — E1–E8
- `.sd/foxotale/ux-ideation.md` — Flutter screen inventory
- Discovery answers 2026-09-28: FastAPI, AWS+Postgres, APAC, recommended pilot sizing

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Definition of Done — Phase 1A

- [x] Problem, scope, stack confirmed
- [x] Architecture diagram + ADR for monolith
- [x] Core entities sketched
- [x] Security context captured
- [x] Pilot scale recommendations recorded
- [ ] User approval to proceed to Phase 1B (contracts)